

WEEK 1 · HANDOUT

Implicature and Presupposition

0 Speaker meaning and the Cooperative Principle

Grice 1975. Logic and conversation.
In *Syntax and Semantics* 3.

DEFINITIONS

Sentence meaning	The conventional contribution of the expressions.
Speaker meaning	What the speaker communicates by choosing them here.
What is said	The contextually fixed content expressed.
What is implicated	Additional inferred content.
Conversational implicature	What is communicated beyond what is said, derived from assumptions about rational, cooperative interaction.
Entailment	A entails B when B holds in every situation in which A holds. Entailments cannot be cancelled and are monotonic.
Cooperative Principle	Assume participants are making appropriate contributions to the purpose of the exchange.
Quality	Try to make your contribution true.
Quantity	Be as informative as required.
Relation	Be relevant.
Manner	Be perspicuous.

(1) A : *Are you coming to the meeting?* B : *I have marking to do.*

Exercise.

(a) For (1), separate what B said from what B implicated.

(b) Name the maxim that licenses the inference, and state the reasoning in one sentence.

1 Implicature by maxim

Context. Each maxim generates a characteristic kind of inference. Quality implicatures often involve apparent falsity treated as purposeful. Quantity implicatures arise when a weaker statement is used in place of a stronger one. Relation implicatures arise when relevance has to be reconstructed. Manner implicatures arise when a speaker is unexpectedly indirect, long, or opaque.

Grice 1975. On flouting: the maxim is violated at the level of what is said, and observed at the level of what is meant.

DEFINITIONS

Flouting	A maxim is violated at the level of what is said and observed at the level of what is meant.
Generalised implicature	Carried by the form itself, absent a special context.
Particularised implicature	Carried only in a particular context, as in (1).

- (2) [someone arrives drenched] *Lovely weather.*
- (3) [a colleague submits a clearly unfinished draft] *This is extremely polished.*
- (4) [a student ignores every instruction] *You followed the brief perfectly.*
- (5) *Pat read the abstract.*
- (6) *Pat produced a sequence of sounds closely corresponding to the score.*

Exercise.

- (a) For each of (2) to (6), state the implicature and the maxim flouted.

- (b) In (2) to (4) the speaker says something they believe false. Say why this does not make them uncooperative.

2 Diagnostics for Implicature

DEFINITIONS

Monotonic	Adding further premises never withdraws a conclusion: if $\Gamma \vdash A$ then $\Gamma \cup \{B\} \vdash A$.
Defeasibility / non-monotonicity	An inference is defeasible when later material can withdraw it.

DIAGNOSTICS

1. **Cancellability.** The inference can be cancelled without contradiction.
2. **Reinforceability.** The inference can be made explicit without redundancy.
3. **Calculability.** It is possible to argue from context plus maxims to the inference.

- (7) *Some of the students passed. In fact I have no idea how many.*
(8) *Some, but not all, of the students passed.*
(9) *Some of the students passed. In fact none of them did.*

Exercise.

- (a) (7) is fine and (9) is not. Say precisely what the difference shows.

- (b) (8) is not redundant. Say what follows about whether *not all* is part of what is said.

- (c) Run all three diagnostics on the implicature of (5). Report any that fail.

3 Scalar implicature and strengthening

Horn 1972. On the semantic properties of logical operators in English.

Context. Scalar implicature is a Quantity-style inference driven by alternatives, and it is a **strengthening** phenomenon: what is communicated is logically stronger than what is said. A speaker uses a weaker term on a scale; the hearer considers stronger alternatives; the strengthened interpretation excludes at least some of them.

DEFINITIONS

Scale	A set of alternative expressions ordered by logical strength.
Alternative	An expression the speaker could have used in place of the one uttered.
Logical strength	A statement <i>A</i> is logically stronger than a statement <i>B</i> iff <i>A</i> entails <i>B</i> but <i>B</i> does not entail <i>A</i> .

DOMAIN	SCALES
Quantifiers	<i>all</i> > <i>most</i> > <i>many</i> > <i>some</i>
Modals	<i>must</i> > <i>should</i> > <i>may</i>
Numerals	<i>three</i> > <i>two</i> > <i>one</i>
Connectives	<i>and</i> > <i>or</i>
Adverbs	<i>always</i> > <i>often</i> > <i>sometimes</i>

A GRICEAN DERIVATION TEMPLATE

1. Utterance: *Some of the students passed.*
2. Stronger alternative: *All of the students passed.*
3. Quantity pressure: the stronger would be more informative.
4. Quality pressure: do not assert what you lack evidence for.
5. Inference: the speaker is not in a position to assert the stronger alternative.
6. Typical strengthening: *not all.*

Exercise.

- (a) Run the template on one item from each row of the table. Give the utterance, the alternative, and the strengthening.

SOURCES OF ORDERING

1. **Entailment.** *All* entails *some*.
2. **Context.** Destinations, rankings, plans, routes.
3. **Discourse.** What would count as a stronger answer to the current question.

Not every scale is ordered by entailment. Travel and progress scales are ordered relative to a goal. Context: travelling north from Cork to Belfast, on the scale {Cork, Wexford, Dublin, Belfast}.

(10) *I made it to Wexford.*

(11) *I have written to every headmaster in Rochdale.*

DEFINITIONS

Presupposition	Content the utterance requires to be already in place, rather than content it puts forward.
Assertion	The content the speaker puts forward as a contribution to the conversation.
Presupposed content	Content treated as background, and preserved where asserted content is reversed or destroyed.
Projection	A presupposition survives embedding under an operator that removes the assertion.
Trigger	A lexical item or construction that introduces a presupposition.
Common ground	The propositions interlocutors treat as mutually accepted for current purposes.
Accommodation	The hearer's silent repair of the context so that the utterance becomes acceptable.

(12) *The prime minister of Switzerland is visiting Dublin.*

Asserted: the prime minister of Switzerland is visiting Dublin.

Presupposed: Switzerland has a prime minister.

PROJECTION UNDER EMBEDDING

1. **Negation.** *The prime minister of Switzerland is not visiting Dublin.*
Asserted content reversed; presupposed content preserved.
2. **Question.** *Is the prime minister of Switzerland visiting Dublin?*
Asserted content destroyed; presupposed content preserved.
3. **Conditional antecedent.** *If the prime minister of Switzerland is visiting Dublin, the meeting will be rescheduled.* Asserted content destroyed; presupposed content preserved.

Hey wait a minute. Background requirements can be challenged in a way asserted content cannot.

(13) *The prime minister of Burkina Faso is visiting Dublin.*

Two replies: *No ... he isn't visiting.* And: *Hey, wait a minute ... Burkina Faso doesn't have a prime minister.*

- (14) Factives. *Berlusconi knows that he is signing the end of Berlusconiism.*
- (15) Aspectual verbs. *China has stopped stockpiling metals.*
- (16) Temporal clauses. *The dude released this video before he went on a killing spree.*
- (17) Manner adverbs. *Jamie ducked quickly behind the wall.*
- (18) Sortally restricted predicates. *Julius is a bachelor.*
- (19) Clefts. *It was Jesus who set me free.*
- (20) Quantifiers. *I have written to every headmaster in Rochdale.*
- (21) Definite descriptions. *The Prime Minister of Trinidad and Tobago stood up and wagged his finger.*
- (22) Names. *The author is Julius Seidensticker.*
- (23) Intonation and focus. *HE set me free.*

Kiparsky & Kiparsky 1970 (factives) ·
 Simons 2001, Abusch 2002
 (aspectual) · Beaver & Condoravdi
 2003 (temporal) · Abbott 2000
 (manner) · Thomason 1972 (sortal) ·
 Delin 1995, Prince 1986 (clefts) ·
 Roberts 1995, Cooper 1983
 (quantifiers) · Strawson 1950
 (definites) · van der Sandt 1992
 (names) · Jackendoff 1972, Geurts &
 van der Sandt 2004 (focus).

DEFINITIONS

The projection problem	How the presuppositions of a complex sentence are determined by those of its parts and by the environments embedding them.
Local support	Earlier material in the sentence already entails the presupposition, so nothing needs to project.

THREE ENVIRONMENTS

1. **Holes** let presuppositions project. *The prime minister of Switzerland is not visiting Dublin.*
2. **Plugs** can block projection to the common ground by confining the presupposition to an embedded context. *Maria thinks that the prime minister of Switzerland is visiting Dublin.*
3. **Filters** make projection depend on other material in the sentence. *If Switzerland has a prime minister, the prime minister of Switzerland is visiting Dublin.*

Exercise. For each item: identify a presupposition candidate; classify the environment as hole, plug or filter; justify in one or two sentences by appealing to projection behaviour.

- (24) *Jamie knows that the prime minister of Switzerland is visiting Dublin.*
- (25) *Jamie believes that the prime minister of Switzerland is visiting Dublin.*
- (26) *A journalist said that the prime minister of Switzerland is visiting Dublin.*
- (27) *Maybe the prime minister of Switzerland is visiting Dublin.*
- (28) *Either Switzerland has no prime minister, or the prime minister of Switzerland is visiting Dublin.*
- (29) *Either the prime minister of Switzerland is visiting Dublin, or the meeting will be cancelled.*
- (30) *Alex thinks that China has stopped stockpiling metals.*

Filtering practice. For each pair, decide whether the presupposition projects from the whole sentence, and justify. Order matters.

- (a) *Switzerland has a prime minister, and the prime minister of Switzerland is visiting Dublin. / The prime minister of Switzerland is visiting Dublin, and Switzerland has a prime minister.*

- (b) *If Switzerland has a prime minister, the prime minister of Switzerland is visiting Dublin. / If the prime minister of Switzerland is visiting Dublin, Switzerland has a prime minister.*
